

KPI Comparison Matrix

Arid-climate resilience indicators

Author

Repository

DOI

Licence

Version

Companion library

Tier system

CORE

EXTENSION

EXPLORATORY

Contents of the matrix

id

name

tier

family

formula

units

domain

source

assumptions

limitations_in_BWk_BSk

regional_extension_rationale

references

How to cite

Citation

Acknowledgement

Plan of work 2026–2029

rix v1.0

icators for residential buildings — South American context

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<https://github.com/<user>/kpi-comparison-matrix> (placeholder — to be created)

10.5281/zenodo.XXXXXXX (placeholder — to be assigned on Zenodo release)

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1.0 · released YYYY-MM-DD

arid-resilience-kpis (Python package implementing the 5 core KPIs)

5 indicators computed on 100 % of cases of the 2026–2029 plan.

7 indicators computed when data quality permits; reinforce the core findings.

5 indicators computed opportunistically; potential contributions subject to availability.

ix sheet

Short stable identifier.

Full descriptive name.

CORE / EXTENSION / EXPLORATORY.

Conceptual family the indicator belongs to.

Formal definition in plain text (Unicode subscripts kept ASCII for portability).

Physical unit; 'dimensionless' when applicable.

What the indicator measures and why it matters.

Primary literature or this work.

Assumptions required for the indicator to be valid.

Documented limitations in arid cold/semi-arid climates of the Köppen-Geiger classification.

Justification for using or extending the indicator in our regional context.

Bibliographic references (semicolon-separated).

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This dataset is the first deliverable of the IEA EBC Annex 80-aligned framework being developed at INAHE-CONICET for arid South America (BWk/BSk). It feeds the matrix of objectives OE1 and OE2 of the 2026–2029 plan and is consumed by the companion Python library `arid-resilience-kpis`.

id	name	tier	family
CCOR	Climate Change Overheating Resistivity	CORE	Annex 80 — Resilience to climate change
DEDT	Energy sensitivity to climate (delta E / delta T)	CORE	Sensitivity
DTDT_MAX	Maximum indoor thermal change rate (dT/dt max)	CORE	Hygrothermal stability
IOD	Indoor Overheating Degree	CORE	Annex 80 — Overheating magnitude
UDH	Unmet Degree Hours during power outage (Passive Survivability)	CORE	Annex 80 — Passive survivability
AIOD_AWC	Building Climate Vulnerability Factor	EXTENSION	Annex 80 — Vulnerability ratio
ITENUATIC	Indoor/outdoor thermal attenuation	EXTENSION	Passive performance

AWD	Ambient Warmness Degree	EXTENSION	Annex 80 — Climate forcing
E_ADAPTIV	Hours of Exceedance over adaptive comfort	EXTENSION	Annex 80 — Adaptive comfort
NARIO_RO	Inter-scenario robustness	EXTENSION	Resilience
SIVE_LOSS	Passive strategy loss under SSP scenario	EXTENSION	Resilience
PHASE_LAG	Thermal phase lag (time shift between outdoor and indoor peaks)	EXTENSION	Passive performance
ACRS	Arid Climate Resilience Score (composite)	EXPLORATORY	Composite
MFORT_PE	Discomfort persistence	EXPLORATORY	Hygrothermal stability

HIHH	Heat Index Hazard Hours	EXPLORATORY	Heat-stress
HYGRO_STABILITY	Joint T-RH-dewpoint stability score	EXPLORATORY	Hygrothermal stability
WET_HOURS	WET-weighted exceedance hours	EXPLORATORY	Annex 80 — Heat-stress weighted

formula	units
$CCOR = (IOD_baseline - IOD_strategy) / \Delta T_climate$	$^{\circ}C \cdot h / ^{\circ}C$
$\Delta E / \Delta T = \text{slope of regression } EUI \sim T_mean \text{ across SSP scenarios and time horizons}$	$kWh \cdot m^{-2} \cdot yr^{-1} / ^{\circ}C$
$\max_t dT_{in}(t)/dt $ computed over the cooling season	$^{\circ}C/h$
$IOD = \sum_z [\sum_t \max(T_{op,z}(t) - T_{comf,z}(t), 0) * dt * occ_z(t)] / \sum_z [\sum_t occ_z(t) * dt]$	$^{\circ}C \cdot h$
$UDH_w = \sum_t \text{in } W \max(T_{op}(t) - T_{threshold}, 0) * dt$, for window W in {24h, 72h, 7d}	$^{\circ}C \cdot h$
$BCVF = IOD / AWD$	dimensionless
$Atten = (\max(T_{in,daily}) - \min(T_{in,daily})) / (\max(T_{ext,daily}) - \min(T_{ext,daily}))$	dimensionless (0 = perfect attenuation, 1 = no attenuation)

AWD = $\sum_t \max(T_{\text{ext}}(t) - T_b, 0) \cdot dt$ (with $T_b = 18^\circ\text{C}$ in the original Annex 80 definition for summer)	$^\circ\text{C}\cdot\text{h}$
HE = $\#\{t : T_{\text{op}}(t) > T_{\text{comf,adapt}}(t) \text{ and } t \text{ is occupied}\}$	hours
$R = \text{std}_{\{m \text{ in } M\}}(\text{IOD}_m)$ for ensemble M of GCMs under fixed SSP and horizon	$^\circ\text{C}\cdot\text{h}$
Loss_SSP = $\text{IOD}_{\text{strategy}}(2080, \text{SSP}) - \text{IOD}_{\text{strategy}}(\text{actual})$	$^\circ\text{C}\cdot\text{h}$
$\phi = t_{\text{peak,indoor}} - t_{\text{peak,outdoor}}$ (median over the cooling season)	hours
ACRS = $\sum_j w_j \cdot \text{normalised_KPI}_j$ with weights w_j from Shannon entropy weighting (Diakoulaki et al., 1995)	dimensionless (0-100 scale after normalisation)
P = mean duration (h) of consecutive runs where $T_{\text{op}} > T_{\text{comf}}$	hours

$HIHH = \# \{ t : HI(T, RH) > HI_threshold \}$	hours
$S = std(T_in) + \alpha * std(RH_in) + \beta * std(Td_in)$ with alpha and beta to be calibrated	dimensionless (composite)
$SET \cdot h = \sum_t \max(SET(t) - 30, 0) * dt$	°C·h

domain	source
Effectiveness of a passive strategy under climate-change-driven temperature shifts	Rahif, Hamdy, Homaei et al. (2022)
Robustness of building energy performance under climate-warming trajectories	Proposed in this work, building on Flores Larsen, Filippín & Barea (2019)
Intra-day dynamic stability of indoor temperature; captures the speed of thermal transients	Proposed in this work
Magnitude of overheating, weighted by zone occupancy	Hamdy, Carlucci, Hoes & Hensen (2017)
Habitability during a sustained power outage triggered during an extreme weather event (EWY)	Sun, Zhang, Zeng, Levinson, Wei & Hong (2021)
Ratio of indoor overheating to ambient warming forcing; measures intrinsic vulnerability of the building	Hamdy, Carlucci, Hoes & Hensen (2017)
Thermal mass effect on damping outdoor amplitude	ISO 13786:2017

Climate-side warming forcing on the building, normalised by base temperature	Hamdy, Carlucci, Hoes & Hensen (2017)
Frequency of exceedance over an adaptive comfort threshold	EN 16798-1:2019; CIBSE TM52/TM59
Inter-model dispersion of building performance across the GCM ensemble	Proposed in this work, following Nik (2016) and Almazroui et al. (2021)
Degradation of a passive strategy under future climate scenarios	Proposed in this work, building on Moazami, Nik, Carlucci & Geving (2019)
Time delay introduced by envelope thermal mass; complements attenuation	ISO 13786:2017; Shaviv, Yezioro & Capeluto (2001)
Single-number summary score over the 5 core KPIs	Proposed in this work, building on Diakoulaki, Mavrotas & Papayannakis (1995)
Severity of discomfort beyond simple counting of exceedance hours	Proposed in this work

Heat stress as Steadman Heat Index exceedance	Steadman (1979); Tao, Sahin, Akbari et al. (2024)
Joint stability of indoor T, RH and dewpoint over a defined period	Proposed in this work
Heat stress weighted by Standard Effective Temperature	Sun et al. (2021); ANSI/ASHRAE 55-2020

assumptions	limitations_in_BWk_BSk
A baseline strategy is defined; Delta_T_climate is the projected mean annual temperature increase under the SSP scenario considered	Requires a locally meaningful baseline; importing a European baseline produces non-comparable results
Linearity of EUI vs T_mean within the SSP × horizon range considered; identical occupancy and HVAC schedules across scenarios	Linearity must be tested empirically; in highly insulated envelopes the response can be non-linear
Time series at 10-min resolution or finer; smoothed by a 3-point moving average to reduce sensor noise	None specific; this indicator is itself an extension specifically motivated by BWk/BSk dynamics
Single comfort threshold T_comf,z; occupancy schedule known per zone; positive part operator; sub-hourly time step dt	T_comf was calibrated on European temperate climates; underestimates intra-day variations characteristic of arid regions with diurnal range > 15 °C
T_threshold = 30 °C (cooling) or 26 °C (more conservative); HVAC unavailable during W; outage triggered at the EWY peak	Threshold 30 °C calibrated to North American climates; low RH typical of arid regions modifies thermal perception (effective temperature differs from dry-bulb)
Linearity of the IOD-AWD relationship; both indicators computed over the same period	Linearity assumption may break in BWk where high diurnal amplitude allows night cooling; BCVF in arid climates may show non-monotonic behaviour
Daily amplitudes; cooling season; HVAC off or in free-running mode	None specific; ISO 13786 is climate-agnostic

T _b = 18 °C; integration over the cooling season	T _b derived from temperate-climate methodologies; in arid hot climates (e.g. summer Mendoza) T _b should be higher to avoid saturation
Adaptive comfort model valid for the climate; occupancy schedule known	European adaptive comfort models have not been validated for arid South American climates
GCM ensemble with at least 8 members; equal weighting (or entropy-weighted variant)	Inter-model dispersion is itself larger over the Andes than over flat domains, so this indicator is essential — not optional — in our region
Same building model; same occupancy schedule; only the EPW file changes between actual and 2080 horizon	None specific
Daily peak detection robust to noise (use smoothed series); free-running mode preferred for evaluation	None specific; well-suited to BWk which presents strong diurnal cycles
Min-max normalisation per KPI before weighting; PCA used as sensitivity check on weights (OECD, 2008)	A composite score always loses information; ACRS must be reported alongside the individual core KPIs, never replacing them
Run-length encoding on the binary series $1\{T_{op} > T_{comf}\}$	None specific; run lengths in arid climates tend to be shorter than in humid climates due to nocturnal cooling, so this indicator captures regime differences

Steadman model calibrated on humid mid-latitude climates	Underestimates heat stress in dry climates; not recommended as primary indicator. Prefer UTCI or WBGT in our region
alpha and beta to be calibrated against occupant comfort surveys; provisional alpha = beta = 1	Coefficients require local calibration
SET model assumes moderate-to-high relative humidity	Very low RH (sometimes <20%) characteristic of the arid regime falls outside the validation envelope of the SET model

regional_extension_rationale	references
Baseline defined as IRAM 11605 Level B (current Argentine residential standard); Delta_T_climate computed from CMIP6 multi-GCM ensemble for the specific city	Rahif et al. (2022) Building and Environment 208, 108599; IEA EBC Annex 80 (Holzer et al., 2024)
Captures degradation of bioclimatic strategies under climate change; particularly relevant in arid regions where cooling demand is projected to rise sharply	Flores Larsen, Filippín & Barea (2019) Energy and Buildings 184; Hamdy et al. (2017)
Conventional aggregated indicators (EUI, HDD/CDD, discomfort hours) are insensitive to intra-day dynamics. In arid climates with diurnal range > 15 °C, dT/dt is a primary discriminator of envelope thermal mass	Cantón, Ganem, Barea & Fernández Llano (2014) Renewable Energy 69; Flores-Larsen, Filippín & Bre (2023) Building and Environment 230
T_comf must be derived from a locally validated adaptive comfort model rather than a fixed European threshold	Hamdy et al. (2017) Building and Environment 122; ASHRAE 55-2020; EN 16798-1:2019
Threshold to be recalibrated using SET (Standard Effective Temperature) for low RH conditions, or alternatively WBGT/UTCI	Sun et al. (2021) Energy and Buildings 252, 111383; Levinson et al. (2019); ANSI/ASHRAE 55-2020
Validate linearity empirically per typology; if non-linear, report the IOD-AWD scatter and the regression residual structure	Hamdy et al. (2017); Attia, Levinson, Ndongo et al. (2021) Energy and Buildings 239
Critical in BWk where outdoor diurnal range frequently exceeds 15 °C; discriminates effectively between adobe (high mass) and lightweight industrialised typologies	ISO 13786:2017 — Thermal performance of building components — Dynamic thermal characteristics; Cantón, Ganem, Barea & Fernández Llano (2014) Renewable Energy 69

T _b regional candidate: 22-24 °C for BWk summer based on local thermal autonomy curves and adaptive comfort models	Hamdy et al. (2017) Building and Environment 122; CIBSE TM52/TM59
Use adaptive comfort models locally validated when available (Pérez-Fargallo et al., 2024 for Chile); otherwise report HE under multiple candidate models with sensitivity analysis	EN 16798-1:2019; CIBSE TM52, TM59; Pérez-Fargallo et al. (2024)
Quantifies the climate uncertainty propagated into the building outcome; reporting only the ensemble mean hides the spread	Nik (2016) Applied Energy 177; Almazroui et al. (2021) Earth Systems and Environment 5(2)
Anchors hypothesis H3 of the plan: passive strategies in BWk lose less effectiveness than in temperate European climates	Moazami et al. (2019) Building and Environment 162; Nik (2016) Applied Energy 177
Strategy of central importance in arid climates: a phase lag of 8-12 h shifts the peak indoor temperature to the cool dawn hours, enabling effective night ventilation	ISO 13786:2017; Shaviv, Yezioro & Capeluto (2001) Energy and Buildings 33; Filippín, Ricard, Flores Larsen & Marek (2022) Energy and Buildings 273
Useful for institutional audiences (IRAM, INTI, IPV) requiring a single score; entropy weighting avoids subjective weighting and is reproducible	Diakoulaki, Mavrotas & Papayannakis (1995) Computers & Operations Research 22; OECD (2008) Handbook on constructing composite indicators
Two buildings with the same exceedance hour count can have very different occupant impact: a cluster of 8 consecutive uncomfortable hours is worse than 8 isolated hours	Carlucci & Pagliano (2012) Building and Environment 53

Reported only for cross-comparability with international literature; flagged as potentially misleading in BWk	Steadman (1979) J. Applied Meteorology 18; Tao et al. (2024) Energy and Buildings 307
In arid climates condensation risk is low but moisture variability still matters for material durability and respiratory comfort	ASHRAE Standard 160-2016 — Criteria for Moisture-Control Design Analysis in Buildings
Compare against UTCI or WBGT; report all three, document which is most consistent with measured discomfort surveys	Sun et al. (2021); ANSI/ASHRAE 55-2020; Gagge, Fobelets & Berglund (1986)

Tier	Colour
CORE	green
EXTENSION	yellow
EXPLORATORY	pink

Operational meaning in the 2026–2029 plan

Computed on 100 % of cases. Forms the committed nucleus of the plan; reported for every house, every typology, every scenario.

Computed where the data quality permits (longer monitoring records, calibrated models). Reinforces core findings; reported for the subset of houses that meet the data-quality threshold.

Computed opportunistically. Constitutes potential additional contributions; reported when a paper or project explicitly requires the indicator and the necessary data is available.